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(54) **RADIOMETER FOR MICROWAVE RECEIVER AND METHOD FOR MEASURING TEMPERATURE OF PROBE THEREOF**

(58) **Field of Classification Search**
CPC H01Q 3/24; G01K 1/026
See application file for complete search history.

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G01K 1/02 (2021.01)

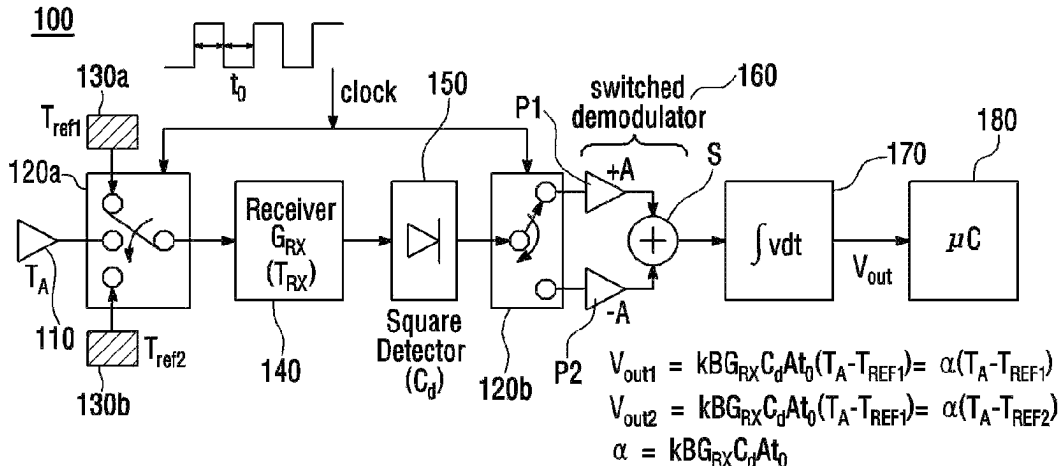
(52) **U.S. Cl.**

CPC **H01Q 3/24** (2013.01); **G01K 1/026** (2013.01)

(57) **ABSTRACT**

A radiometer for a microwave receiver according to an embodiment of the present invention comprises: an antenna line through which radiation power corresponding to the temperature of a probe is received from an antenna; a first reference rod having a first reference temperature fixed in advance; a second reference rod having a second reference temperature fixed in advance; a first microwave switch for selectively switching the antenna line, the first reference load, and the second reference load; a second microwave switch switched with the same clock as the first microwave switch; a demodulator for multiplying a signal output from the second microwave switch by a predetermined gain value and outputting same; an integrator for integrating the signal output from the demodulator; and a controller for measuring the temperature of the probe on the basis of a signal output from the integrator.

6 Claims, 2 Drawing Sheets



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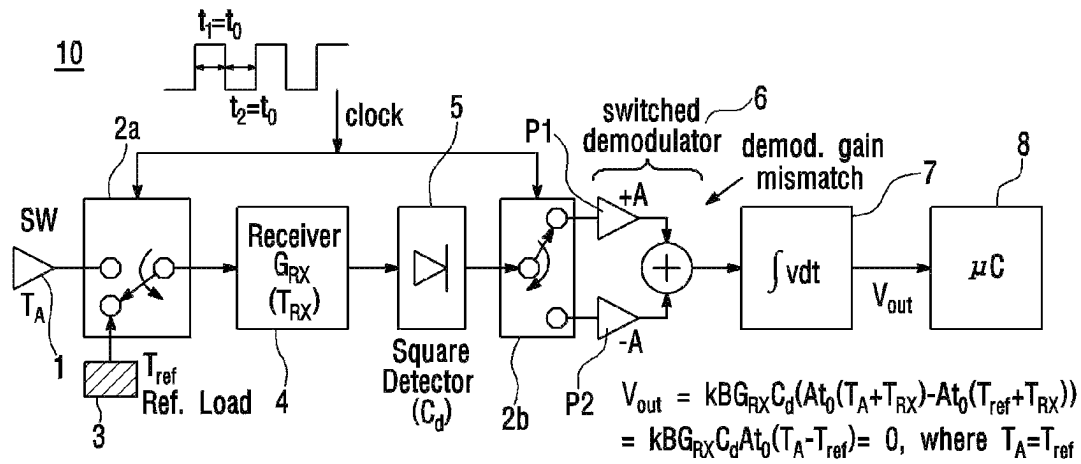


FIG. 1

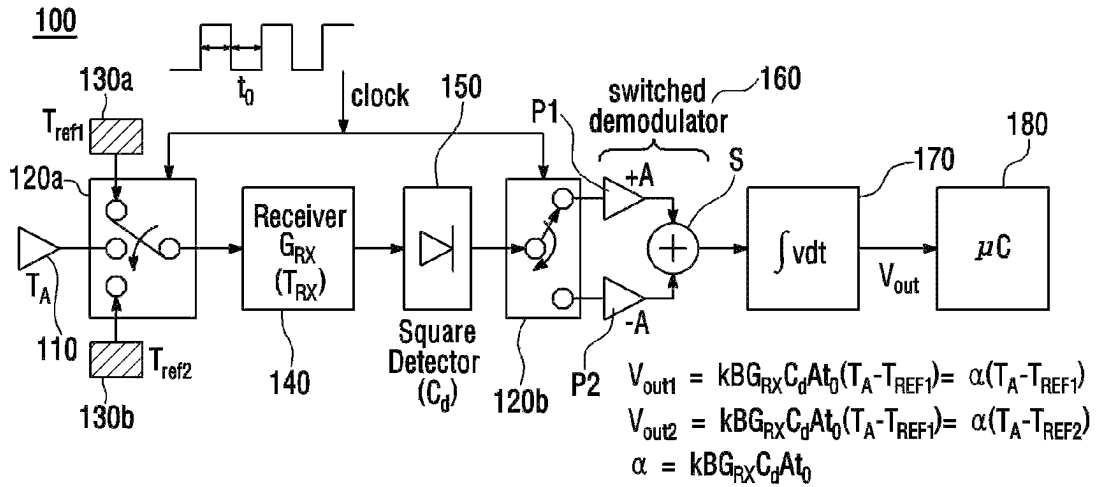


FIG. 2

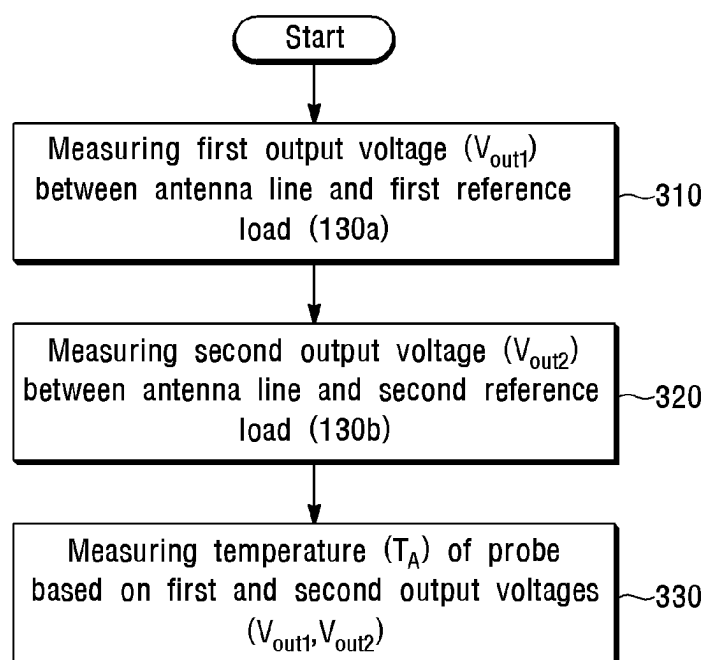


FIG. 3

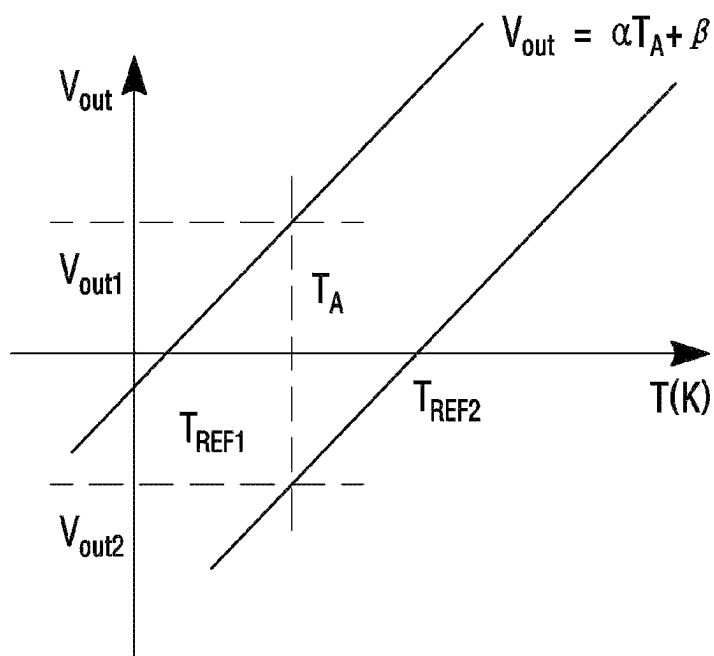


FIG. 4

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RADIOMETER FOR MICROWAVE RECEIVER AND METHOD FOR MEASURING TEMPERATURE OF PROBE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This U.S. non-provisional patent application is a U.S. national stage application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2022/002083, filed Feb. 11, 2022, which claims priority to Korean Patent Application No. 10-2021-0052224, filed on Apr. 22, 2021. The disclosures of the aforementioned priority applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present invention relates to a radiometer for a microwave receiver and a method for measuring a temperature of a probe thereof, and more particularly, to a radiometer for a microwave receiver, which is capable of easily and accurately measuring a temperature of a probe without varying a reference a reference temperature of a reference rod and a method for measuring a temperature of a probe thereof.

BACKGROUND ART

According to Planck's law of blackbody radiation, light radiated from a material, a wavelength of a radio wave, and an amount of a component may be expressed as a function of a temperature. In a microwave frequency range (300 MHz to 300 GHz), according to Rayleigh-Jeans law, a radiation signal is expressed by [Mathematical equation 1] below to have a value proportional to a square of a frequency f and a temperature T . Thus, radiation power P_n in a frequency bandwidth BW received by an antenna is expressed by [Mathematical equation 2] below. When the radiation power received through a microwave receiver by using the [Mathematical equation 2], i.e., noise power is amplified and finally measured, a temperature proportional thereto may be measured, and this is called as a radiometer.

$$B^p(f) = \frac{2hf^2}{c^2} (1 + hf/kT - 1)^{-1} = \frac{2f^2kT}{c^2} \quad [\text{Mathematical equation 1}]$$

(k : Boltzmann constant, c : speed of light in vacuum, T : absolute temperature, f : frequency)

$$P_n = kT\Delta f \quad [\text{Mathematical equation 2}]$$

The radiometer is classified into: a total power radiometer that simply amplifies received radiation power (noise) and measures an output; and a Dicke radiometer having an excellent temperature resolution unlike the total power radiometer.

FIG. 1 is a view for explaining a typical Dicke radiometer and a reception operation principle thereof.

Referring to FIG. 1, in the typical Dicke radiometer, microwave switches **2a** and **2b** and a demodulator **6** are switched with the same clock. A temperature of an object to be detected is received by a first microwave switch **2a** as radiation power corresponding to ' T_A ' through an antenna **1**.

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Also, reference power corresponding to ' T_{ref} ' of a reference load **3** (Ref. Load) is received by the first microwave switch **2a** for comparison with T_A . The radiation power corresponding to T_A or the reference power corresponding to T_{ref} is input to the receiver **4** according to switching of the first microwave switch **2a**. The receiver **4** processes an input power signal and outputs same to a detector **5**, and an output of the detector **5** is input to the second microwave switch **2b**.

When the first microwave switch **2a** is connected to the antenna **1**, the second microwave switch **2b** connects the detector **5** to a first amplifier **P1** of the demodulator **6**. On the other hands, when the microwave switch **2a** is connected to a reference load **3**, the second microwave switch **2b** connects the detector **5** to a second amplifier **P2** of the demodulator **6**. Here, the first amplifier **P1** and the second amplifier **P2** have the same gain and opposite polarities $+A$ and $-A$. Also, ' C_d ' is a detector constant of the detector **5**.

The demodulator **6** outputs the output of the first amplifier **P1** and the second amplifier (**P2**) to an integrator **7**, and the output voltage V_{out} of an integrator **7** is input to the controller **8**.

The radiation power corresponding to T_A during a half cycle T_1 and the reference power corresponding to the T_{ref} during a remaining half-cycle T_2 is input to the integrator **7**. The output voltage V_{out} of the integrator **7** becomes ' $kBG_{RX}C_D A_{ro}(T_A - T_{ref})$ ' when passes through the integrator **7**. Here, when T_A equals to T_{ref} , the output voltage V_{out} becomes '0'. Since the output voltage of the typical Dicke radiometer is proportional to $(T_A - T_{ref})$ unlike the total power radiometer, the typical Dicke radiometer may have T_{RX} of the receiver **4** that is not affected, be insensitive to change of a gain G_{RX} of the receiver **4**, and also obtain T_A to be measured by controlling the T_{ref} .

However, the typical Dicke radiometer controls ' T_{ref} ' for measuring of a temperature of a probe to be measured, i.e., ' T_A ' corresponding to the radiation power input through the antenna **1**, and this controlling is substantially difficult and complicated to be implemented. Specifically, the typical Dicke radiometer finds a moment of matching with the T_A by varying the equivalent temperature T_{ref} or resistance of the reference rod **3**, and the moment is when the output voltage V_{out} of the integrator **7** becomes '0'. However, the moment in which the varied T_{ref} is hard to find, and the variable T_{ref} is not easy to create. Therefore, a radiometer capable of measuring the temperature of the probe more easily and accurately is required.

DISCLOSURE OF THE INVENTION

Technical Problem

The present invention provides a radiometer for a microwave receiver, which is capable of easily and accurately measuring a temperature of a probe to be measured and a method for measuring a temperature of a probe thereof.

Technical Solution

Embodiments of the present invention provide a radiometer for a microwave receiver, including: an antenna line through which radiation power corresponding to a temperature of a probe is received from an antenna: a first reference rod having a first reference temperature fixed in advance; a second reference rod having a second reference temperature fixed in advance; a first microwave switch for selectively switching the antenna line, the first reference load, and the second reference load; a second microwave switch switched

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with the same clock as the first microwave switch; a demodulator for multiplying a signal output from the second microwave switch by a predetermined gain value and outputting same; an integrator for integrating a signal output from the demodulator; and a controller for measuring the temperature of the probe based on a signal output from the integrator.

In other embodiments of the present invention, a method for measuring a temperature of a probe of a radiometer for a microwave receiver includes: a first switching step in which the first microwave switch is alternately switched between the antenna line and the first reference load according to the clock; a second switching step in which the first microwave switch is alternately switched between the antenna line and the second reference load according to the clock; and a measuring step of measuring the temperature of the probe based on a first output voltage output from the integrator by the first switching step and a second output voltage output from the integrator by the second switching step.

Advantageous Effects

The radiometer for the microwave receiver according to the embodiment of the present invention may easily and accurately measure the temperature of the probe.

Particularly, the radiometer may measure the temperature of the probe through the easy and simple calculation by using the two reference temperatures fixed at the specific values without varying and controlling the reference temperature of the reference rod in comparison with the typical Dicke radiometer.

DESCRIPTION OF DRAWINGS

FIG. 1 is a view for explaining a typical Dicke radiometer and a reception operation principle thereof.

FIG. 2 is a view illustrating a radiometer for a microwave receiver according to an embodiment of the present invention.

FIG. 3 is a flowchart representing a method for measuring a temperature T_A of a probe of a radiometer for a microwave receiver according to an embodiment of the present invention in FIG. 2.

FIG. 4 is a graph showing the method for measuring the temperature T_A of the probe of the radiometer for the microwave receiver according to an embodiment of the present invention in FIG. 2.

MODE FOR INVENTION

Hereinafter, the present invention may be best appreciated by referring to the following detailed description of preferred embodiments with reference to the accompanying drawings. In the drawings, like reference numerals refer to like elements throughout, and redundant descriptions will be omitted. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention. In the following description, the technical terms are used only for explain a specific exemplary embodiment while not limiting the present invention. In addition, the principle, viewpoint, and embodiments of the present invention and all detailed descriptions of specific embodiments should be understood to be intended to include the structural and functional equivalents of the matter.

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FIG. 2 is a view illustrating a radiometer for a microwave receiver according to an embodiment of the present invention.

Referring to FIG. 2, the radiometer for the microwave receiver according to an embodiment of the present invention includes two reference rods **130a** and **130b** in comparison with a typical Dicke radiometer illustrated in FIG. 1, and a first microwave switch **120a** of the radiometer electrically connects one of an antenna **110**, a first reference load **130a**, and a second reference load **130b** to a receiver **140** according to switching.

More specifically, the radiometer according to an embodiment of the present invention includes an antenna **110**, microwave switches **120a** and **120b**, first and second reference loads **130a** and **130b**, a receiver **140**, a detector **150**, a demodulator **160**, an integrator **170**, and a controller **180**.

The microwave switches **120a** and **120b** include a first microwave switch **120a** and a second microwave switch **120b**. The first microwave switch **120a** and the second microwave switch **120b** are switched with the same clock.

The first microwave switch **120a** is electrically connected to an antenna line and each of the first and second reference rods **130a** and **130b**. Also, the first microwave switch **120a** is electrically connected to the receiver **140**. Here, the antenna line represents a path electrically connected to the antenna **110**. Components except for the antenna **110** may be disposed on the line.

The first microwave switch **120a** may include a first terminal electrically connected to the antenna line, a second terminal electrically connected to the first reference load **130a**, a third terminal electrically connected to the second reference load **130b**, and a fourth terminal electrically connected to the receiver **140**.

The first microwave switch **120a** electrically connects one of the antenna **110**, the first reference load **130a**, and the second reference load **130b** to the receiver **140** according to switching. When the first microwave switch **120a** electrically connects the antenna line to the receiver **140**, radiation power corresponding to a temperature T_A of a probe is transmitted from the antenna **110** to the receiver **140** through the antenna line. On the other hand, when the first microwave switch **120a** electrically connects the first reference load **130a** to the receiver **140**, first reference power corresponding to a first reference temperature T_{ref1} is input to the receiver **140**, and when the first microwave switch **120a** electrically connects the second reference load **130b** to the receiver **140**, second reference power corresponding to a second reference temperature T_{ref2} is input to the receiver **140**.

The receiver **140** may be a super heterodyne receiver having a bandwidth B and a total power gain G_{RX} .

The receiver **140** processes a signal input from the first microwave switch **120a**. For example, the receiver **140** may predetect one of a power signal input by switching of the first microwave switch **120a**, radiation power corresponding to the temperature T_A from the antenna **110**, first reference power corresponding to the first reference temperature T_{ref1} from the first reference load **130a**, and second reference power corresponding to the second reference temperature T_{ref2} from the second reference load **130b**.

The detector **150** may be a square-law detector that outputs a signal proportional to a square of an amplitude of the signal input from the receiver **140**, and include a semiconductor diode.

The second microwave switch **120b** is connected between the detector **150** and the demodulator **160**. The second microwave switch **120b** electrically connects the detector

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150 to a first amplifier **P1** or a second amplifier **P2** according to switching. The second microwave switch **120b** has a first output terminal connected to the first amplifier **P1** and a second output terminal connected to the second amplifier **P2**.

The demodulator **160** includes the first amplifier **P1**, the second amplifier **P2**, and a combiner **S**. The first amplifier **P1** and the second amplifier **P2** have the same gain while having opposite polarities $+A$ and $-A$ and are each electrically connected to the second microwave switch **120b**. Outputs of the first amplifier **P1** and the second amplifier **P2** are added and output by the combiner **S**.

The integrator **170** accumulates signals input from the demodulator **160** for a predetermined time to transmit a predetermined output voltage to the controller **180**.

The controller **180** may measure the temperature T_A of the probe based on the output voltage from the integrator **170**. A specific method for measuring the temperature T_A of the probe will be described in detail below.

Hereinafter, a method for measuring the temperature T_A of the probe by the radiometer for the microwave receiver according to an embodiment of the present invention in FIG. 2 will be described with reference to FIG. 3.

FIG. 3 is a flowchart representing the method for measuring the temperature T_A of the probe of the radiometer for the microwave receiver according to an embodiment of the present invention in FIG. 2.

The method for measuring the temperature T_A of the probe of the radiometer for the microwave receiver according to an embodiment of the present invention uses the first reference temperature T_{ref1} and the second reference temperature T_{ref2} , which are fixed at specific values.

Referring to FIG. 3, the method for measuring the temperature T_A of the probe of the radiometer for the microwave receiver according to an embodiment of the present invention includes: a step **310** of measuring a first output voltage V_{out1} between the antenna line and the first reference load **130a**; a step **320** of measuring a second output voltage V_{out2} between the antenna line and the second reference load **130b**; and a step **330** of measuring the temperature T_A of the probe based on the measured first and second output voltages.

The step **310** of measuring the first output voltage V_{out1} between the antenna line and the first reference load **130a** will be described. When as the first and second microwave switches **120a** and **120b** are switched with the same clock, the first microwave switch **120a** alternately switches the antenna **110** and the first reference load **130a**, and the second microwave switch **120b** alternately switches the first amplifier **P1** and the second amplifier **P2** of the demodulator **160**, the first output voltage V_{out1} of the integrator **170** is expressed by [Mathematical equation 3] below.

$$V_{out1} = kBG_{RX}C_dAt_0(T_A - T_{ref1}) = \alpha(T_A - T_{ref1}) \quad [\text{Mathematical equation 3}]$$

Here, K is a Boltzmann constant, B is a bandwidth of the receiver, G_{RX} is a gain of the receiver, C_d is a detector constant, A is a gain of the first and second amplifier, and t_0 is a half-cycle time of clock.

Next, the step **320** of measuring the second output voltage V_{out2} between the antenna line and the second reference load **130b** will be described. When as the first and second microwave switches **120a** and **120b** are switched with the same clock, the first microwave switch **120a** alternately switches the antenna **110** and the second reference load

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130b, and the second microwave switch **120b** alternately switches the first amplifier **P1** and the second amplifier **P2** of the demodulator **160**, the second output voltage V_{out2} of the integrator **170** is expressed by [Mathematical equation 4] below.

$$V_{out2} = kBG_{RX}C_dAt_0(T_A - T_{ref2}) = \alpha(T_A - T_{ref2}) \quad [\text{Mathematical equation 4}]$$

Lastly, the step **330** of measuring the temperature T_A of the probe based on the measured first and second output voltages V_{out1} and V_{out2} will be described. A ratio of the first output voltage V_{out1} of the [Mathematical equation 3] and the second output voltage V_{out2} of the [Mathematical equation 4] is shown in [Mathematical equation 5] below.

$$\frac{V_{out1}}{V_{out2}} = \frac{(T_A - T_{ref1})}{(T_A - T_{ref2})} \quad [\text{Mathematical equation 5}]$$

The [Mathematical equation 5] is summarized by the temperature T_A of the probe to be measured and expressed into [Mathematical equation 6] below.

$$T_A = \frac{[T_{ref1}(V_{out2}/V_{out1}) - T_{ref2}]}{[(V_{out2}/V_{out1}) - 1]} \quad [\text{Mathematical equation 6}]$$

When the temperature T_A of the probe is calculated by the [Mathematical equation 6], unnecessary α may be removed. That is, the temperature T_A of the probe may be easily and accurately measured based on the two reference temperatures T_{ref1} and T_{ref2} fixed at specific values and the measured two first and second output voltages V_{out1} and V_{out2} .

FIG. 4 is a graph showing the method for measuring the temperature T_A of the probe of the radiometer for the microwave receiver according to an embodiment of the present invention in FIG. 2.

As shown in the graph of FIG. 4, although the T_A may be measured when an inclination α of a linear function on V_{OUT} is known, the T_A may not be measured by measuring only one reference temperature. Thus, by using the radiometer according to an embodiment of the present invention in FIG. 2, the temperature T_A of the probe may be accurately measured from the two reference temperatures T_{ref1} and T_{ref2} and the output voltages V_{out1} and V_{out2} of each of the two reference temperatures T_{ref1} and T_{ref2} . The radiometer according to an embodiment of the present invention may be unnecessary to control and vary the reference temperature of the reference load to measure the temperature T_A of the probe and easily and accurately measure the temperature T_A of the probe regardless of characteristics of inner component of the radiometer, i.e., the bandwidth B of the receiver **140**, a detector constant C_d of the detector **150**, and the gains and polarities of the first and second amplifiers **P1** and **P2**.

On the other hand, the above-described method for measuring the temperature of the probe of the radiometer for the microwave receiver may be realized in the form of a program command implemented through various computer units and recorded in computer readable media. The computer readable recording media may include a program command, a data file, and a data structure or a combination thereof. The program command recorded in the above-described media may be particularly designed and configured for the present invention or may be well-known and

used to those skilled in computer software. The computer readable recording media include e.g., magnetic media such as a hard disk, a floppy disk, and a magnetic tape, optical media such as CD-ROM and DVD, magneto-optical media such as a floptical disk, and a hardware device that is specially configured to store and perform a program command such as ROM, RAM, and a flash memory. The program command includes, e.g., a high language code that is executed by a computer using an interpreter as well as a machine code such as a compiler. The hardware device may be configured to operate one or more software modules to perform operations of the present invention, and vice versa.

Although a preferred embodiment of the present invention has been disclosed, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims. It is also understood that the terms used herein are merely descriptive, rather than limiting, and that various changes may be made without departing from the scope and spirit of the invention.

DESCRIPTION OF REFERENCE NUMERAL

110: Antenna

120a: First microwave switch

120b: Second microwave switch

130a: First reference rod

130b: Second reference rod

140: Receiver

150: Detector

160: Demodulator

170: Integrator

180: Controller

The invention claimed is:

1. A radiometer for a microwave receiver, comprising:
 - an antenna line through which radiation power corresponding to a temperature of a probe is received from an antenna;
 - a first reference rod having a first reference temperature fixed in advance;
 - a second reference rod having a second reference temperature fixed in advance;
 - a first microwave switch for selectively switching the antenna line, the first reference load, and the second reference load;
 - a second microwave switch switched with the same clock as the first microwave switch;
 - a demodulator for multiplying a signal output from the second microwave switch by a predetermined gain value and outputting same;
 - an integrator for integrating a signal output from the demodulator; and
 - a controller for measuring the temperature of the probe based on a signal output from the integrator.
2. The radiometer of claim 1, wherein the signal output from the integrator comprises:
 - a first output voltage based on a feature in which the first microwave switch alternately switches the antenna line and the first reference load; and

a second output voltage based on a feature in which the first microwave switch alternately switches the antenna line and the second reference load, wherein the temperature of the probe is measured based on the first and second reference temperatures and the first and second output voltages.

3. The radiometer of claim 2, wherein the temperature of the probe is determined by the Mathematical equation below

$$T_A = \frac{[T_{ref1}(V_{out2}/V_{out1}) - T_{ref2}]}{[(V_{out2}/V_{out1}) - 1]}$$

Here, T_A is the temperature of the probe, T_{ref1} is the first reference temperature, T_{ref2} is the second reference temperature, V_{out1} is the first output voltage, and the V_{out2} is the second output voltage.

4. The radiometer of claim 1, further comprising a receiver and a detector disposed between the first microwave switch and the second microwave switch, wherein the demodulator comprises:
 - a first amplifier connected to a first output terminal of the second microwave switch;
 - a second amplifier connected to a second output terminal of the second microwave switch; and
 - a combiner configured to add output signals of the first amplifier and the second amplifier,
 wherein the first amplifier and the second amplifier have the same gain and opposite polarities.

5. A method for measuring the temperature of the probe of the radiometer for the microwave receiver of claim 1, comprising:

- a first switching step in which the first microwave switch is alternately switched between the antenna line and the first reference load according to the clock;
- a second switching step in which the first microwave switch is alternately switched between the antenna line and the second reference load according to the clock; and
- a measuring step of measuring the temperature of the probe based on the first output voltage output from the integrator by the first switching step and the second output voltage output from the integrator by the second switching step.

6. The method of claim 5, wherein the temperature of the probe is determined by the Mathematical equation below

$$T_A = \frac{[T_{ref1}(V_{out2}/V_{out1}) - T_{ref2}]}{[(V_{out2}/V_{out1}) - 1]} <$$

Here, T_A is the temperature of the probe, T_{ref1} is the first reference temperature, T_{ref2} is the second reference temperature, V_{out1} is the first output voltage, and the V_{out2} is the second output voltage.

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